INCH-POUND

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12 October 1998
SUPERSEDIING
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10 March 1976

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPES 7YP1, 7YP2, AND 7YP7

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Electrostatic deflection and focus, square face, five beam.

DIMENSIONS AND PIN CONNECTIONS: See figure 1.

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Eb1	Eb2	Eb3	ed	Rg	Zd	Ehk	Eb3/Eb2
Unit:	V	V dc	V dc	V dc	V dc	v	Meg	Meg	V dc	Ratio
Maximum:	6.9	-200	1,650	3,850	7,700	1,250	1.5	1.0 see note 1	±180	2.0 see note 2
Minimum:	5.7	0	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -
Test conditions: see note 3	6.3	Adjust focus		2,000	4,000	- - -	- - -	1.0	- - -	2.0

GENERAL:

Qualification - Required.

Prior to shipment, 7YP2 tubes shall be electrically burned-in on a 100 percent basis. These tubes shall be operated at standard operating conditions with 5 microamperes direct current applied to each gun for a period of 48 hours. During this 48 hour period, the tube shall be turned off for 30 minutes (allowed to cool) and then turned on to continue burn-in at least 4 times. Following burn-in, each tube shall be tested in accordance with the following MIL-STD-1311 tests:

Method	Test	Method	Test
5201	Voltage breakdown	5241	Grid-cutoff voltage
5206	Gas “cross”	5251	Grid No. 1 leakage current
5223	Modulation	5251	Anode No. 2 leakage current
5231	Spot displacement (leakage)	5201	Electrode current (Anode No. 1)

Those tubes which fail any of these tests shall not be shipped.

Visual inspections shall be made to ensure that all JAN 7YP2 CRT's are failure free from arcing for continuous and intermittent operational testing during all manufacturing and acceptance testing.

Pressure test method 1141 shall be performed, except 30 pounds per square inch (PSI) absolute pressure shall be applied in lieu of 45 PSI absolute, on a 100 percent basis prior to shipment for the JAN 7YP2 tube only. Only those tubes meeting the acceptance criteria for this test shall be shipped.

When a P2 CRT screen is specified , the phosphors shall be in compliance with Electronic Industries Alliance TEPAC Publication, TEP 116-B, ‘Optical Characteristics of Cathode Ray Tubes.’

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TABLE I. Testing and inspection.

Inspection	Method	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
Conformance inspection, part 1							
Voltage breakdown	5201	- - -		- - -	- - -	- - -	- - -
Voltage breakdown (electrostatic types)	5201	- - -		- - -	- - -	- - -	- - -
Gas “cross”	5206	6, 11	Ib3 = 50 µA dc	- - -	- - -	- - -	- - -
Bulb, screen and faceplate quality	5106	- - -		- - -	- - -	- - -	- - -
Light output	5221	11 11	P1 screen; Ib3 = 50 µA P2 screen; Ib3 = 50 µA, for 7YP2 only	Light Light	25 25	- - - - - -	ftL ftL
Modulation	5223	- - -	Ib3 = 50 µA dc	ΔEc1	- - -	55	V dc
Spot position (electrostatic deflection)	5231	8		- - -	- - -	16	mm
Spot displacement (leakage)	5231	- - -		- - -	- - -	10	mm
Grid cutoff voltage	5241	- - -		Ec0	-90	-50	V dc
Grid No. 1 leakage current	5251	- - -		Ic1	- - -	3	µA dc
Anode No. 2 leakage current	5251	- - -		Ib2	- - -	10	µA dc
Conformance inspection, part 2							
Screens Type P2 Type P7 Type P7	5221	 11, 13 11, 13 11, 13		 D1(1) D1(5) G5:1	 360 400 4	 - - - - - - - - -	 cB cB cB
Heater current	1301	- - -		Ib	540	660	mA
Electrode current (anode No. 1)	5201	- - -		Ib1	-15	+10	µA dc
Electrode current (cathode)	5201	- - -	Ib3 = 50 µA dc	Ik	- - -	1,000	µA dc
Base alignment (electrostatic types)	5101	- - -	3D4; base key	- - -	- - -	- - -	- - -
Trace and bulb alignment (rectangular electrostatic types)	5101	- - -	1D2 trace and bulb wall with contact button	- - -	- - -	3	Degrees
Side terminal alignment (electrostatic types)	5101	- - -	3D4	- - -	- - -	- - -	- - -

See notes at end of table I.

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TABLE I. Testing and inspection -Continued.

Inspection	Method	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
Conformance inspection, part 2 - Continued							
Side terminal and base alignment	5101		Base key, collar index pin	- - -	- - -	- - -	- - -
Neck and base alignment (electrostatic types)	5101	- - -		- - -	- - -	- - -	- - -
Line width “A” (electrostatic deflection)	5226	- - -	Ib3 = 25 μA dc	Width	- - -	0.65	mm
Line width “B” (electrostatic deflection)	5226	7	Ib3 = 25 μA dc	Width	- - -	0.80	mm
Focusing voltage at Ib = 25 μA direct current	5246	- - -		Eb1	450	650	V dc
Deflection factor	5248	9	1D2	DF	68	84	V dc/in.
Deflection factor	5248	10	3D4	DF	27	37	V dc/in.
Heater-cathode leakage current	5251	- - -		Ihk	- - -	15	μA dc
Angle between traces	5101	- - -		- - -	88	92	Degrees
Trace alignment	- - -	12		- - -	- - -	- - -	- - -
Base pin solder depth	1111	- - -	Sample size shall be 1 percent of lot (lot = 1 month’s production)	- - -	- - -	0 failures	- - -
Secureness of base, cap or insert	1101	15		- - -	- - -	- - -	- - -
Permanence of marking	1105	- - -	For qualifications only.	- - -	- - -	- - -	- - -
Conformance inspection, part 3							
Life test	- - -	14	Group C; t = 500 hours; Ib3 = 25 μA dc	- - -	- - -	- - -	- - -
Life-test end points:	- - -						
Modulation	5223	- - -	Ib3 = 40 μA dc	ΔEc1	- - -	55	V dc
Line width “A”	5226	- - -	Ib3 = 18 μA dc	Width	- - -	0.65	mm
Line width “B”	5226	- - -	Ib3 = 18 μA dc	Width	- - -	0.80	mm
Pressure (implosion)	1141	5		- - -	- - -	- - -	- - -
Vibration	5111	5	For qualification only.	Width	- - -	1.0	mm

See notes at end of table I.

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TABEL I. Testing and inspection -Continued.

Inspection	Method	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
Conformance inspection, part 3 - Continued							
Direct-interelectrode capacitance	1331	4, 5	k to all g1 to all D1 to D2 D3 to D4 D1 to all D2 to all D3 to all D4 to all	Ck Cg1 C1D2 C3D4 CD1 CD2 CD3 CD4	- -	6.5 7.0 2.7 3.2 8.1 8.0 7.8 7.8	pF pF pF pF pF pF pF pF
Neck and bulb alignment (electrostatic types)	5101	5		- - -	- - -	- - -	- - -
Stray light emission (conventional types)	5216	5	Eb2 = 3,850 V dc; Eb3 = 7,700 V dc	- - -	- - -	- - -	- - -
Cathode illumination	5216	5	For qualification only	- - -	- - -	- - -	- - -
Deflection-factor uniformity	5248	5	1D2 3D4	- - - - - -	- - - - - -	2 5	% %
Interaction factor	5250	5		- - -	- - -	$10x10^{-6}$	in./V dc
Shock	5115	5	For qualification only	- - -	- - -	- - -	- - -
Base material insulating quality	1216	5	For qualification only	- - -	- - -	- - -	- - -

NOTES:

1. The deflection electrode circuit resistance should be 1 MΩ or less and approximately equal, otherwise beam shift at high drives can be expected. Higher resistance values up to 5 MΩ may be used for low-beam current operation.
2. This tube is designed for optimum performance when operating at an Eb3/Eb2 ratio of 2.0. Operation at other ratios of Eb3/Eb2 may result in changes in deflection uniformity and pattern distortion.
3. All tests except direct-interelectrode capacitance, vibration and interaction factor, shall be made on each unit separately. Each deflection plate shall be connected to A2 through a resistance of 1 MΩ.
4. All other electrodes contained within the envelope are to be tied together.
5. This test shall be performed during the initial production and once each succeeding 12-calender months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an Acceptance Level of 6.5. The regular “12 calendar month” double sampling plan shall be reinstated after three consecutive samples have been accepted.
6. The pattern size shall be adjusted to 1.5 inches in the limited scan direction.
7. Due to limited scan of 3D4 plates, the length of scan in the 3D4 direction shall be limited to ±1 inch from the undeflected spot position. Line width “B” shall be measured at ± 0.75 inch from the undeflected spot position.
8. From the spot positions as indicated on figure 1.

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TABLE I. Testing and inspection - Continued.

9. The deflection in this direction covers the minimum useful screen width following the bulb contour.

10. The deflection in this direction is limited to ±0.75 inch minimum from the undeflected spot position.

11. This test to be performed at the conclusion of the holding period. This test applies only when P1 CRT screen phosphors are specified.

12. Corresponding traces of each unit are to be within 2 degrees of each other.

13. This test is not required unless specified in the purchase order or contract.

14. The tube manufacturer (7YP2 only) shall provide the user with two years of service life warranty commencing with shipment from the tube manufacturer. The second year of this warranty shall be pro-rated. Other terms and agreements regarding this warranty (i.e., pro-rating, handling of defectives) shall be negotiated at the time of purchase. The appropriate applicable documentation shall be provided by the tube manufacturer and this documentation shall accompany the tube. The user will be responsible for completing and returning the documentation to the manufacturer.

15. This test shall be performed on 100 percent basis and torque shall be 20 inch-pounds (7YP2 only). Immersion test for qualification only.

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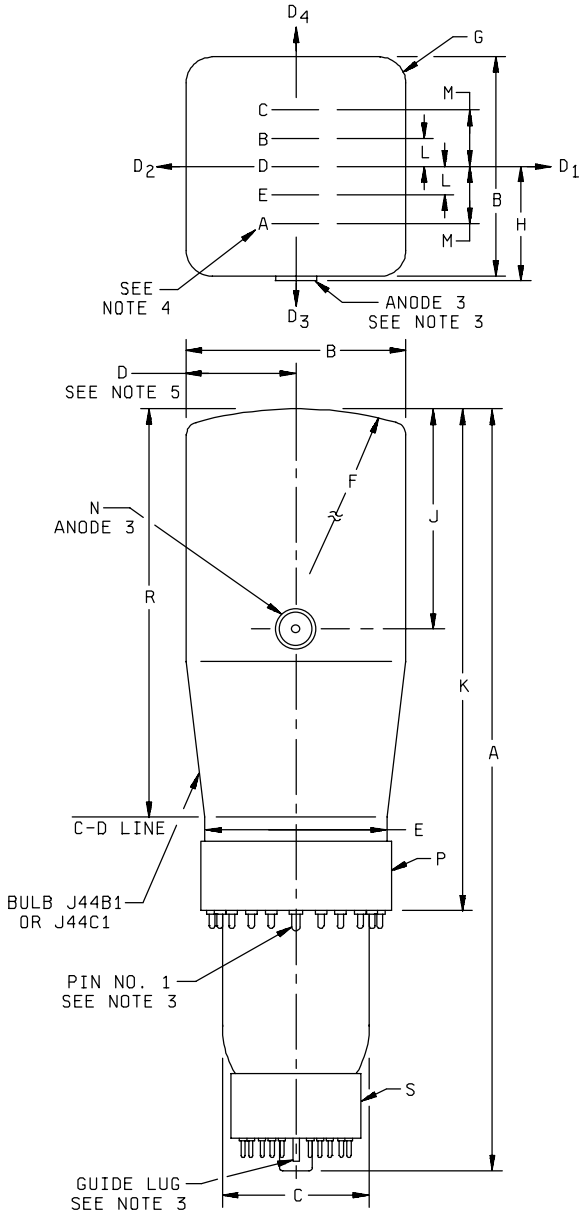

Dimensions in inches with metric equivalents (mm) in parentheses		
Ltr	Minimum	Maximum
Conformance inspection, part 2		
A	18.125 (460.38)	18.875 (479.42)
B	5.438 (138.13)	5.531 (140.49)
C	3.688 (93.68)	3.813 (96.85)
D	2.250 (57.15)	
E	4.344 (110.34)	4.468 (113.49)
H		2.797 (71.04)
J	5.125 (130.18)	5.375 (136.52)
K	12.125 (307.98)	12.375 (314.32)
Conformance inspection, part 3 (See note 5)		
N P S	Bulb contact: J1-22 Base, collar: A22-101 Base, B25-139 (See note 2)	
Reference dimensions (See note 1)		
F	40.00 (1016.0)	
G	0.75 (19.0)	
L	0.75 (19.0)	
M	1.50 (38.1)	
R	10.00 (254.0)	

NOTES:
1. These dimensions are for information and are not required for inspection purposes.
2. For pin alignment use JEDEC gage GB25-1.
3. Bulb contact J1-22, pin number 1 of base A22-101, and guide lug of base B25-139 shall each align with 3D4 trace within 10°.
4. Beam A shall center on position A as shown within 0.312 R. This requirement shall also apply to beams B, C, D, and E and their corresponding positions.
5. Useful screen radius.
6. Lancaster Glass bulb outline A-4314 or equivalent.

FIGURE 1. Outline drawing of electron tube types 7YP1, 7YP2 and 7YP7.

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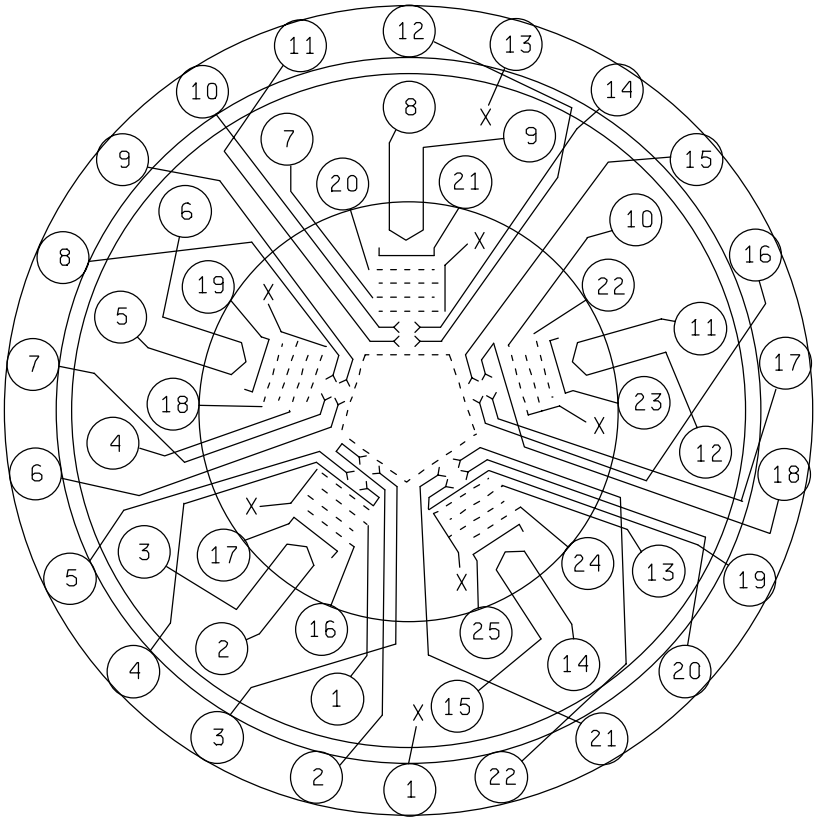

Pin no.	22 Pin collar base Element	Beam
1	Accelerator	
2	Deflector D2	A
3	Deflector D1	A
4	Deflector D3	A
5	Deflector D4	A
6	Deflector D1	B
7	Deflector D3	B
8	Deflector D4	B
9	Deflector D2	B
10	Deflector D3	C
11	Deflector D1	C
12	Deflector D2	C
13	Accelerator	
14	Deflector D4	C
15	Deflector D1	D
16	Deflector D2	D
17	Deflector D4	D
18	Deflector D3	D
19	Deflector D4	E
20	Deflector D3	E
21	Deflector D2	E
22	Deflector D1	E

Pin no.	25 Pin base Element	Beam
1	Focusing electrode	A
2	Heater	A
3	Heater	A
4	Focusing electrode	B
5	Heater	B
6	Heater	B
7	Focusing electrode	C
8	Heater	C
9	Heater	C
10	Focusing electrode	D
11	Heater	D
12	Heater	D
13	Focusing electrode	E
14	Heater	E
15	Heater	E
16	Grid number 1	A
17	Cathode	A
18	Grid number 1	B
19	Cathode	B
20	Grid number 1	C
21	Cathode	C
22	Grid number 1	D
23	Cathode	D
24	Grid number 1	E
25	Cathode	E

FIGURE 1. Outline drawing of electron tube types 7YP1, 7YP2 and 7YP7 - Continued.

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Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AV, CR4, ME
Navy - AS, CG, MC, OS
Air Force - 11, 17, 99

Preparing activity:
DLA - CC

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